C–N–200a
August 8, 1961

SUPERSEDING
Fed. Spec. C–N–200
February 17, 1953
(See 6.2)

FEDERAL SPECIFICATION

NEAT'S-FOOT OIL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Neat's-foot oil covered by this specification is suitable for the treatment of leather.

1.2 Classification.

1.2.1 *Types.* The neat's-foot oil shall be of the following types as specified (see 6.1):

Type I—Non-sulfonated.

Type II—Sulfonated.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitations for bids, unless otherwise specified, form a part of this specification to the extent specified herein.

Federal Specifications:

KK–L–311—Leather, Methods of Sampling and Testing.

TT–I–558—Ink, Marking Stencil, Opaque, for Nonporous Surfaces (Metals, Glass).

PPP–C–96—Cans, Metals, 28 Gage and Lighter.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 536—Soap and Soap Products (Including Synthetic Detergents) Sampling and Testing.

Fed. Test Method Std. No. 791—Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Methods Booklet of the American Leather Chemists' Association.

(Application for copies should be addressed to the University of Cincinnati, Cincinnati 21, Ohio.)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Freight Classification Rules Committee, 1 Park Avenue, New York 16, New York.)

3. REQUIREMENTS

3.1 Type I. The nonsulfonated neat's-foot oil, shall be a clear liquid, free from sediment and admixture of other oils. The oil shall be free from rancidity and the odor shall be nonobjectionable. The oil when tested as specified in section 4 shall conform to the requirements specified in table I.

TABLE I.—*Physical and chemical properties*

	Minimum	Maximum	Test paragraph
Color (N.P.A.)	—	2.5	4.2.1.1
Specific gravity, 15.5°/15.5°C.	0.912	0.920	4.2.1.2
Pour point (°C.)	—	-4	4.2.1.3
Free fatty acid, as oleic acid, percent	—	2.0	4.2.1.4
Saponification number, mg. KOH per gram	192	200	4.2.1.5
Iodine number (Wijs)	—	80	4.2.1.7

3.2 Type II. The oil entering into the manufacture of the sulfonated neat's-foot oil shall conform to the requirements specified in table II.

TABLE II.—*Requirements for oil*

	Minimum	Maximum	Test paragraph
Iodine number (Wijs)	77	95	4.2.2.3
Saponification number, mg. KOH per gram	190	210	4.2.2.4
Free fatty acid, percent	—	2.0	4.2.2.1
Pour point, (°C.)	1.0	6.0	4.2.2.5

3.2.1 *Finished sulfonated neat's-foot oil.* The finished sulfonated neat's-foot oil shall be a homogeneous solution and when tested as specified in section 4 shall conform to the requirements specified in table III. The oil shall be suitable for use in making water emulsions for application to leather when tested as specified in 4.2.2.6.

TABLE III.—*Properties of finished sulfonated neat's-foot oil*

Characteristic	Minimum, percent	Maximum, percent	Test method paragraph
Moisture	24	26	4.2.2.1
Total fatty oil	68	—	4.2.2.1
Unsaponifiable matter	—	2.0	4.2.2.1
Ash	—	5.0	4.2.2.1
Sulphur trioxide (combined with oil)	3.5	5.5	4.2.2.1
Total alkali	—	1.5	4.2.2.1
Free fatty acid as oleic	17	25	4.2.2.1
Stability of 10 per cent emulsion in water (hours)	24	—	4.2.2.6
pH of 10 percent emulsion in water	5.0	6.5	4.2.2.2

3.3 Workmanship. Neat's-foot oil shall be clean, clear, homogeneous, free from sediment and shall conform to the quality of levels established herein.

4. SAMPLING INSPECTION AND TEST PROCEDURES

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to

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assure supplies and services conform to prescribed requirements.

4.1.1 *Sampling for inspection and acceptance.* Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Military Standard MIL–STD–105, except where otherwise indicated herein.

4.1.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.1.1.2 *Inspection of the end item.*

4.1.1.2.1 *Visual examination of the filled container.* Visual examination of the filled container shall be in accordance with the classification of defects indicated below. The lot size shall be expressed in units of filled containers of neat's-foot oil. The sample unit shall be one filled container. The inspection level shall be L–8 with an acceptable quality level (AQL) of 2.5 defects per hundred units.

Examination	Defect
Container	Not in container specified.
Condition of container	Seepage or leakage of contents.
	Dented, cracked, or crushed affecting serviceability.
	Improper closure.
	Indication of rust.
Labeling or marking	Omitted, illegible, incorrect, incomplete or not in accordance with requirements.

4.1.1.2.2 *Examination of the end item for defects in color, odor, and workmanship.* The sample unit for this examination shall be one filled container. The lot size shall be expressed in units of filled containers. The inspection level shall be L–4. The acceptable quality level shall be 2.5 defects per hundred units.

Examination	Defect
Color	Varies from specified color.
Odor	Objectionable odor.
	Rancid.
Workmanship	Liquid not clear.
	Liquid not clean.
	Presence of sediment.
	Liquid not homogeneous.

4.1.1.2.3 *Examination of the end item for net contents of filled containers.* The sample unit for this examination shall be one filled container. The lot size shall be expressed in units of filled containers. The inspection level shall be L–4 except that a minimum of 5 sample units will be selected for examination.

Examination	Defect
Net contents or volume	Average volume or weight less than specified.

4.1.1.2.4 *Examination of packaging, packing, and marking for delivery.* Visual examination of packaging, packing and marking shall be performed in accordance with the classification of defects below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be L–4 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

Examination	Defect
Markings (exterior and interior container)	Incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective affecting serviceability.
Workmanship (as applicable)	Inadequate application of components such as: incomplete closure of case liners, container flaps; loose strapping inadequate stapling; bulged or distorted containers.
Contents	Number per container is more or less than required.
Weight	Gross/net weight exceeds requirements.

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4.1.2 *Testing of the end item.* The end item shall be tested for the characteristics of section 3 herein. The lot size for testing shall be expressed in pints, quarts, or gallons of neat's-foot oil, as specified. The sample unit shall be one filled container. The sample for testing shall consist of a one quart composite taken from all the unit containers sampled. The inspection level for testing shall be L–2 ,except that a minimum of 5 sample units shall be selected for composite testing. Duplicate determinations for compliance with the requirements shall be made on the composite sample. The average of the two determinations shall comply with the requirements. The requirement noted in 3.1 that the neat's-foot oil be free from a mixture of other oil will be satisfied by acceptance of a Certificate of Compliance from the supplier.

4.2 Test procedures.

4.2.1 *Type I.*

4.2.1.1 *Color and appearance.* The thoroughly mixed sample shall be extra lemon pale or lighter in color and shall be determined in accordance with method 102.6 of Federal Test Method Standard No. 791.

4.2.1.2 *Specific gravity.* The specific gravity shall be determined by any method accurate to the third decimal place.

4.2.1.3 *Pour point.* The pour point shall be determined in accordance with method 201.8 of Federal Test Method Standard No. 791.

4.2.1.4 *Free fatty acid.* The free fatty acid shall be determined in accordance with method 5105.3 of Federal Test Method Standard No. 791, except 100 ml. of neutral ethyl alcohol shall be used.

4.2.1.5 *Saponification number.* The saponification number shall be determined in accordance with method 5401.7 of Federal Test Method Standard No. 791. Hold titrated solution for use in the unsaponifiable matter limitation test (see 4.2.1.6).

4.2.1.6 *Unsaponifiable matter limitation test.* Add distilled water to the titrated solution of the saponification number determination (see 4.2.1.5) to approximately 10 times the volume of the titrated solution. Cool to room temperature (20° to 25° C.). The cooled solution shall be clear.

4.2.1.7 *Iodine number.* The iodine number for nonsulfonated oil shall be determined in accordance with method 1101.2 of Federal Test Method Standard No. 536, except that the procedure for preparation of sample shall be omitted.

4.2.2 *Type II.*

4.2.2.1 The tests specified in table IV shall be conducted in accordance with the Methods Booklet of the American Leather Chemists Association.

TABLE IV.—*Test methods*

Tests	Test methods
Moisture	H–40
Total fatty oil	H–45
Unsaponifiable matter[1]	H–47
Ash	H–22
Sulphur trioxide	H–42
Total alkali	H–49
Free fatty acids	H–52

[1] Method shall be as specified except that 40 mls of water shall be added to the mixture after saponification, then the alcohol evaporated prior to extraction with petroleum ether.

4.2.2.2 *pH. 10 percent emulsion.* The acidity, pH. 10 percent emulsion shall be determined in accordance with method 6621 of Federal Specification KK–L–311.

4.2.2.3 *Iodine number.* The iodine number for sulfonated oil shall be determined in accordance with method 1101.2 of Federal Test Method Standard No. 536, except that the procedure for preparation of sample shall be omitted.

4.2.2.4 *Saponification.* The saponification number shall be determined in accordance with method 5401.7 of Federal Test Method Standard No. 791.

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4.2.2.5 *Pour point.* The pour point shall be determined in accordance with method 201.8 of Federal Test Method Standard No. 791.

4.2.2.6 *Stability of emulsion.* Measure with a stoppered, graduated cylinder, 10 ml. of sulfonated neat's-foot oil and make up to 100 ml. with distilled water at 70° F. Mix thoroughly and allow to stand for 24 hours. No oil should separate in that time.

5. PREPARATION FOR DELIVERY

(For Civil agency procurement, Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.3).

5.1.1 *Level A.* One pint, one quart or one gallon, (see 6.3) of type I or type II neat's-foot oil shall be furnished in square or oblong cans conforming to type V, class 4 of Federal Specification PPP–C–96. Each can shall be provided with an innerseal. Filled one-pint cans shall be packaged in accordance with the interior packaging requirements of the Appendix of Federal Specification PPP–C–96.

5.1.2 *Level C.* Type I or type II neat's-foot oil shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.3).

5.2.1 *Level A.* Neat's-foot oil of the same type, packaged as specified in 5.1, shall be packed in accordance with the exterior packing requirements for overseas shipment as specified in the Appendix of Federal Specification PPP–C–96.

5.2.2 *Level B.* Neat's-foot oil of the same type, packaged as specified in 5.1, shall be packed in accordance with the exterior packing requirements for domestic shipment as specified in the Appendix of Federal Specification PPP–C–96.

5.2.3 *Level C.* Neat's-foot oil of the same type, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Labeling.* The label data shall be durably and legibly marked in accordance with Military Standard MIL–STD–129, on the can's body by printing, stenciling or lithographing with a non-corrosive, weather-resistant, waterproof black ink conforming to the requirements of Federal Specification TT–I–558. The label data shall be as follows:

Fed. Spec. No.
Nomenclature
Type
Net contents
Date of pack

5.3.2 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.3 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 *Type I.* Nonsulfonated neat's-foot oil covered by this specification is intended for use in the treatment of leather.

6.1.2 *Type II.* Sulfonated neat's-foot oil covered by this specification is suitable for application to leather in the form of a water emulsion for the purpose of softening and preserving the leather as in the sterilization and rebuilding

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of shoes and in the treatment of used leather to prevent cracking.

6.2 This specification incorporates the requirements of Military Specification MIL–N–1056A, dated 5 August 1952.

6.3 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type required.

(*c*) Inspection responsibility, methods, and tests (see section 4).

(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(*e*) Size of can required (see 5.1.1).

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Oil, Neat's-foot, not otherwise indexed by name.

Carload minimum weight 30,000 pounds.

Motor:

Oil, Neat's-foot, not otherwise indexed.

Truckload minimum weight (W) 30.6 pounds, (W) subject to Rule 34, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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OTHER INTEREST:

M O

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